The Interaction Tax: How Communication Erases Diversity in Multi-Agent Teams

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Abstract

Multi-agent LLM systems combine models from different families on the premise that collaboration outperforms any single model. Recent work shows single models can match multi-agent systems under equal budgets [Tran and Kiela, 2026, Xu et al., 2026, Li et al., 2024], and that unconstrained multi-agent teams consistently underperform their best member [Jarrett et al., 2025]. Yet existing evaluations conflate two distinct factors: whether using diverse models helps, and whether letting agents interact helps. We disentangle these on eleven scientific optimization tasks, evaluating ten configurations built from three model families (Claude, GPT, and Gemini) under hidden scoring where agents never see the final evaluator.

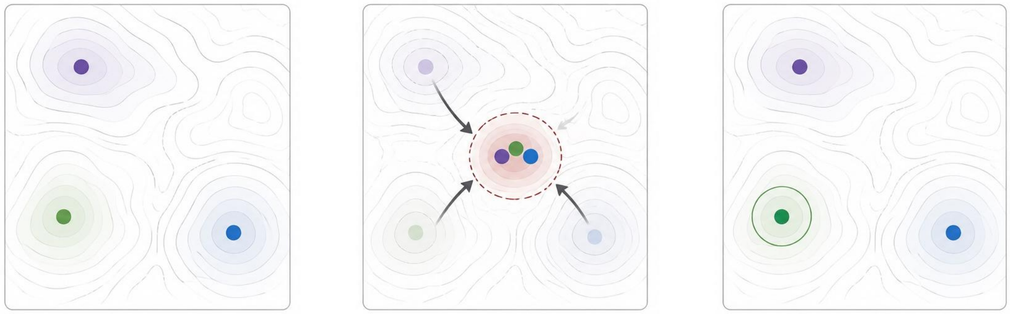
Using diverse models creates an advantage because each family may find solutions the others miss. However, this advantage disappears once agents communicate. Additionally, in every configuration where agents read and revise each other's solutions, their outputs converge within a single round. We call this phenomenon **the interaction tax**.

Whether critique helps depends on verifiability. When errors are checkable, critique raises success rates from 0% to 47–73%. On the other hand with open optimizations, critique degrades solutions 57% of the time. Effective multi-agent systems should therefore preserve proposal independence to protect model diversity and reserve iterative critique for tasks with verifiable error signals. Multi-agent interaction can erase the diversity it was designed to exploit.

1 Introduction

Scientific optimization rewards the single best solution found, not the average quality of attempts. It is also a domain where model diversity should matter most because open problems lack known optimal solutions, different model families approach them with different inductive biases, and there is no ground-truth oracle to anchor convergence toward a correct answer. A team of scientists is epistemically stronger than its members when each brings genuinely different knowledge. Once they share intermediate work, their conclusions become correlated. The wisdom-of-crowds effect [Surowiecki, 2004] breaks down because correlated errors compound rather than cancel [Lorenz et al., 2011]. The same logic applies to multi-agent AI.

Multi-agent LLM systems are built on the assumption that diverse models, working together, outperform any one of them. Recent work applies multi-agent frameworks to coding [Hong et al., 2023, Wu et al., 2023], mathematical reasoning [Du et al., 2024, Chen et al., 2025], and scientific discovery [Romera-Paredes et al., 2024, Georgiev et al., 2025]. Different model families encode different inductive biases and tend to find structurally different solutions. For example, on our benchmark Claude solves Circle Packing perfectly (Q=1.0) while GPT-4o scores 0.519, but on Erdős Overlap GPT-4o leads (Q=0.710) while Gemini scores zero. The premise of multi-agent collaboration is that combining these complementary strengths covers more of the search space than any single



(a) Independent proposals (b) Interaction drives convergence (c) Selection preserves diversity

Figure 1: The interaction tax on optimization tasks. Different model families (Claude, GPT-4o, Gemini) find structurally different solutions, each reaching a different local optimum. When agents interact via debate, critique, or revision, their proposals converge and the diversity is lost. When agents generate independently and the best solution is selected without interaction, diversity is preserved and the strongest solution survives.

model. Whether that premise holds depends on whether the diversity in agents' proposals survives the interaction step designed to leverage it (Figure 1). Recent evidence suggests it does not. Jarrett et al. [Jarrett et al., 2025] show that self-organizing LLM teams consistently underperform their best member, losing up to 37.6% through "integrative compromise." However, their study uses same-model teams with different roles, leaving open whether the loss stems from the interaction itself or from the lack of genuine model diversity. We test both factors independently.

Testing this on scientific optimization requires a fair comparison. Naïve comparisons attribute gains to collaboration that may come from repeated sampling. A better comparison samples the single model as many times as the multi-agent system samples its models. We use this approach across nine scientific optimization tasks and two constraint satisfaction tasks, with ten agent configurations. Optimization results are scored by a hidden evaluator that the agents cannot query during search. Constraint-task comparisons use a common feasibility evaluator.

Our central finding is that interaction has opposite effects depending on whether agents share a backbone. To measure this, we introduce Marginal Interaction Gain (MIG), which compares each configuration against the same agents running independently without communication. On same-model teams, interaction consistently helps. On diverse teams, it consistently hurts, with proposals converging within a single round of full output exchange. We call this the *interaction tax*. Mixture-of-Agents (MoA) is the only architecture that escapes, because its proposers never see each other's outputs. Task structure modulates this pattern. On constraint satisfaction tasks where violations are checkable and locally fixable, iterative critique raises feasibility from 0% to 47–73% (Section 4), showing that convergence can be constructive when feedback is verifiable.

We make three contributions. (1) MIG as a controlled comparison measure that holds agent count and sample budget constant, revealing that interaction flips from positive to negative once backbones become diverse. A 2×2 factorial isolates the mechanism. (2) Preliminary evidence of a task-structure boundary where interaction helps on constraint satisfaction (two tasks) but hurts on open-ended optimization. (3) A hidden-evaluator benchmark methodology for scientific optimization that prevents agents from gaming the scoring function, with visible and hidden rankings diverging on three of nine tasks. These results suggest that effective multi-agent systems should preserve proposal independence when using diverse models and separate visible feedback from final scoring.

2 Related Work

Agentic methods. A large variety of agentic methods emerged in recent works. Self-Refine [Madaan et al., 2023] and Reflexion [Shinn et al., 2023] iterate within a single model. Tree of Thoughts [Yao et al., 2024] structures single-model search as branching deliberation. Multi-agent debate frameworks

[Du et al., 2024, Liang et al., 2023], MAgICoRe [Chen et al., 2025], and MoA [Wang et al., 2025] add cross-agent critique or aggregation. AutoGen [Wu et al., 2023] and MetaGPT [Hong et al., 2023] orchestrate multi-agent conversations but evaluate using visible scorers or LLM judges. None uses a hidden deterministic verifier or isolates the interaction step from agent selection.

Evolutionary approaches. FunSearch [Romera-Paredes et al., 2024], Evolution through Large Models [Lehman et al., 2023], and AlphaEvolve [Georgiev et al., 2025] embed LLMs inside evolutionary loops for mathematical discovery. Four of our benchmark tasks (Circle Packing, Flat Polynomials, Difference Bases, Erdős Overlap) are adapted from the AlphaEvolve problem suite with smaller parameterizations and independent scoring implementations.

Single vs. multi-agent comparisons. Tran and Kiela [Tran and Kiela, 2026], Xu et al. [Xu et al., 2026], and Li et al. [Li et al., 2024] show single agents match or exceed multi-agent systems under equal token budgets or that agent-count scaling helps only for majority-vote tasks. Jarrett et al. [Jarrett et al., 2025] find that self-organizing LLM teams consistently underperform their best member, attributing the loss to "integrative compromise" where teams average expert and non-expert views. Our work differs in three ways. We use genuinely different model families rather than same-model teams with different roles. We introduce a hidden-evaluator split that reveals when visible and hidden rankings diverge. We also identify a task-structure boundary where interaction helps (verifiable constraints) rather than finding it uniformly harmful.

Ensemble diversity theory. The ensemble diversity decomposition [Krogh and Vedelsby, 1994] shows that ensemble error equals mean member error minus a diversity term measuring disagreement among members. When members are independent, diversity is maximized. When members communicate and converge, diversity shrinks toward zero and ensemble error approaches the mean member error. The interaction tax we observe is a structural instance of this. Communication between agents increases inter-model correlation [Hansen and Salamon, 1990, Lakshminarayanan et al., 2017], which directly reduces the diversity term. MIG should therefore be negative whenever interaction increases correlation more than it reduces individual error, and positive when individual error reduction dominates.

3 Methods

3.1 Setup

Each task is an optimization problem with a fixed answer format. A deterministic verifier scores solutions without human judgment. A configuration is a fixed setup specifying how many agents interact, in what order, and whether they see each other's outputs. Each configuration receives the task description, a budget vector, and one or more backbone LLMs, and returns its solution. The budget vector caps token usage at T=200,000, wall-clock time at W=600s, evaluator CPU at C=30s per call, and total visible-evaluator calls at K=25 (i.e., agents may query the visible scoring function at most 25 times during a single run to evaluate candidate solutions). All configurations share identical caps, so any performance differences reflect solution quality rather than compute expenditure.

Each task has two evaluators. The visible evaluator is queryable during search. The hidden evaluator is held out and runs once on the best solution after the configuration terminates, so agents never observe their hidden-evaluator score during the run. Table 1 lists both evaluators for each task. The hidden evaluator serves the same role as a held-out test set in supervised learning. It prevents configurations from overfitting to the visible scoring function and ensures that reported gains reflect genuine solution quality rather than evaluator exploitation. Without this split, a configuration that queries the visible evaluator more effectively can appear superior even if its solutions are no better on the underlying problem. On three optimization tasks (Difference Bases, MaxCut, Molecule QED) visible and hidden configuration rankings diverge (Spearman $\rho < 1.000$). These are open problems without known optimal solutions [Applegate et al., 2006, Bickerton et al., 2012], so it is unsurprising that two different evaluation criteria can rank methods differently. All optimization comparisons therefore use hidden-evaluator scores. On constraint tasks, we report feasibility rates from the visible evaluator because the analysis compares configurations under the same evaluator rather than testing for visible-hidden divergence.

Let s denote the raw hidden-evaluator score for a given solution. We map it to $Q(s) = (s - b) \,/\, (r - b)$, clipped to $[0, 1]$, where b is the trivial-solution baseline score (e.g., random partition for MaxCut) and

Table 1: Eleven benchmark tasks. Dir. is scoring direction (max or min). The visible evaluator is queryable during search. The hidden evaluator scores the final solution once. Constraint tasks test feasibility under iterative critique. Their hidden evaluators are listed for completeness, but the constraint analysis in Section 4 uses visible-evaluator feasibility only.

Task	Domain	Dir.	Visible evaluator	Hidden evaluator
Optimization tasks				
MaxCut $G(200, 0.3)$	Graph	max	80% edge subset	Full edge set
Circle Packing n=20	Geometry	max	Overlap tol. 10^{-6}	Overlap tol. 10^{-12}
Difference Bases	Combin.	min	100% contiguous coverage	95% robust coverage
Flat Poly. deg. 50	Analysis	min	100K uniform grid points	500K pts + adversarial peak
TSP-100	Routing	min	Exact cities	Cities perturbed ±5 units
LJ n=41	Chemistry	min	Standard LJ potential	LJ + Axilrod–Teller 3-body
Erdős Overlap	Analysis	min	N=1000 discretisation	N=10,000 + adversarial shift
Molecule QED	Chemistry	max	Plain QED score	QED $\times$ synthesizability
TSP-50	Routing	min	Exact cities	Cities perturbed ±3 units
Constraint satisfaction tasks				
Knapsack-50	Combin.	max	Original weights	Perturbed weights ±3
3AP-Free-100	Combin.	max	Subset size (0 if AP found)	Same

Table 2: Agent configurations. All operate under the same budget vector (T=200K, W=600s, C=30s, K=25). HPE = Hierarchical Planner-Executor. Two MoA ablations (same-model proposers, no synthesis) are used in the 2×2 analysis.

Configuration	Role	Mechanism	Ref.
Single-Shot	Control	One model, one attempt, no iteration.	—
Best-of-N (N=8)	Control	8 independent samples, return visible-best.	—
Self-Refine	Control	Same model generates, critiques, revises for 3 rounds.	Madaan et al. [2023]
VGS	Control	Evolutionary loop (pop. 4, gen. 5, elite 2) with visible scores as fitness.	Romera-Paredes et al. [2024]
Homo-Chain	Multi	4 same-model agents in sequence, each revises prior output. Visible-best returned.	This work
Cross-Chain	Multi	3-agent chain (Claude → GPT-4o → Gemini). Visible-best returned.	This work
MAgICoRe	Multi	Solver → reviewer → refiner, 2 rounds.	Chen et al. [2025]
Debate	Multi	2 agents propose, exchange critiques (2 rounds), synthesizer merges.	Du et al. [2024]
HPE	Multi	Planner decomposes, executors solve subproblems, integrator assembles.	Wang et al. [2023], Hong et al. [2023]
MoA	Multi	3 proposers generate independently, 1 synthesizer combines.	Wang et al. [2025]

r is the best known published result for each task. We write $Q_{\text{hidden}}(p,\, t)$ for the mean Q-score of configuration p on task t, averaged over random seeds.

3.2 Tasks

The benchmark contains nine optimization tasks and two constraint satisfaction tasks (Table 1). Four optimization tasks are adapted from the AlphaEvolve problem suite [Georgiev et al., 2025] with smaller parameterizations (Circle Packing n=20 vs. 26, Flat Polynomials degree 50 vs. 69) and independent scoring implementations. The remaining tasks draw on classical problems in combinatorics, routing [Applegate et al., 2006], Lennard-Jones cluster optimization [Wales and Doye, 1997], and drug-likeness scoring [Bickerton et al., 2012]. Each optimization task pairs a visible evaluator with a hidden evaluator that runs once on the best solution and is never queried by agents. Three optimization tasks (Difference Bases, Erdős Overlap, and Molecule QED) produce meaningful configuration spread and account for most of the signal when comparing configurations. On two tasks (MaxCut and LJ-n=41) every configuration scores near zero ($Q \approx 0$), meaning no model finds a non-trivial solution. These tasks provide no signal for comparing methods but are retained for completeness. The two constraint satisfaction tasks test whether the interaction mechanism changes under tasks with checkable local violations.

3.3 Agent Configurations

We test ten configurations (Table 2). Best-of-N (N=8) serves as the independent-sampling baseline. Homo-Chain and Cross-Chain share the sequential-revision structure but differ in backbone composition (same vs. diverse) and chain length (4 vs. 3 agents). Together they form one of the five configuration families compared in MIG analysis. MoA is the only configuration where proposers never read each other's outputs.

3.4 Comparison Measures

Marginal Epistemic Gain (MEG). MEG measures whether a configuration beats the best single-agent baseline (positive = yes). $\mathrm{MEG}(p,t) = Q_{\mathrm{hidden}}(p,t) - \max_{c \in \mathcal{C}} Q_{\mathrm{hidden}}(c,t)$, where $\mathcal{C} = \{\text{Self-Refine}, \text{Best-of-}N, \text{VGS}\}$. Aggregate MEG averages across nine optimization tasks with $10{,}000$ bootstrap CIs. A jackknife correction of $+0.006$ adjusts for finite-sample bias in $\max$. No configuration changes sign.

Marginal Interaction Gain (MIG). MIG measures whether interaction helps (positive) or hurts (negative) relative to running the same agents independently. $\mathrm{MIG}(p,t) = Q_{\mathrm{hidden}}(p,t) - Q_{\mathrm{hidden}}(\mathrm{parallel}(A), t)$, where $\mathrm{parallel}(A)$ runs each agent in configuration p once independently (no interaction, no shared context) and returns the maximum score among them. For a 3-agent diverse configuration, this means one Claude, one GPT-4o, and one Gemini sample. For a 3-agent same-model configuration, this means three independent samples from the same backbone. Both baselines use the same number of samples, so MIG differences are not confounded by baseline strength. MIG is computed separately for same-model and diverse-model configurations. Aggregate MIG averages over eight tasks (TSP-50 excluded for incomplete coverage).

3.5 Backbones and seeds

The three backbones are Claude Sonnet 4 [Anthropic, 2025], GPT-4o [OpenAI, 2024], and Gemini 2.5 Flash [Google DeepMind, 2025], all via OpenRouter. In the core benchmark, same-model configurations use Claude throughout. Within-backbone follow-ups in Section 4 and Appendix use Gemini. Each (task, configuration) cell uses five seeds in the main benchmark. The $2{\times}2$ factorial uses more seeds to achieve adequate power for coefficient estimation, specifically ten per arm on Erdős and MolQED and thirty on DiffBases where per-seed variance is higher ($N{=}198$ total after excluding two DiffBases runs that failed budget validation). Within-backbone comparisons use $n{=}15$ based on the lesson that $n{=}5$ produced a false positive on MolQED that dissolved at $n{=}15$.

4 Results

Per-model coverage. No single backbone dominates. On the six tasks where backbone Best-of-N scores clearly diverge, each model is the top scorer on at least one task (Table 9, Appendix). Claude leads on Circle Packing ($Q{=}1.000$), GPT-4o leads on Erdős ($Q{=}0.710$), and Gemini leads on Difference Bases ($Q{=}0.610$) and Flat Polynomials. Any single-backbone team faces catastrophic failure. Gemini scores $Q{=}0.000$ on Erdős and GPT-4o scores $Q{=}0.007$ on Flat Polynomials. A diverse team provides coverage without requiring prior knowledge of which model fits which task.

The diversity coefficient. A $2{\times}2$ factorial crosses backbone diversity (same vs. diverse proposers) with synthesis (MoA synthesis vs. best-score selection) on three tasks with sufficient Q-score variance (Difference Bases, Erdős, Molecule QED, $N{=}198$ runs, task-stratified bootstrap). The remaining six tasks show near-zero variance across all four arms, precluding coefficient estimation.

Table 3: $2{\times}2$ factorial estimates. Diversity is the only factor whose CI excludes zero. 95% CI from 10,000 task-stratified bootstrap resamples, $N{=}198$.

Term	Estimate	95% CI
Intercept (same + selection)	$+0.181$	$[+0.146,\ +0.220]$
Diversity	$+0.195$	$[+0.125,\ +0.262]$
Synthesis	$+0.044$	$[-0.023,\ +0.113]$
Diversity $\times$ Synthesis	$+0.035$	$[-0.103,\ +0.172]$

The diversity coefficient is $+0.195$ with a CI entirely above zero (Table 3, Figure 2), positive in all 10,000 bootstrap resamples. The synthesis coefficient is $+0.044$ with a CI that straddles zero. The result replicates across synthesizers (Appendix H). Per-task cell means (Appendix L) show Erdős contributes $+0.48$, DiffBases $+0.126$, and MolQED $+0.026$.

Leave-one-out analysis (Appendix G) shows Erdős drives the coefficient, and removing it drops the estimate to $+0.052$ (CI $[-0.013,\ +0.117]$, borderline). Inter-model error correlations (Appendix N) show near-zero ρ on Erdős and DiffBases but $\rho {\approx} +0.43$ on TSP-100. Diversity adds information

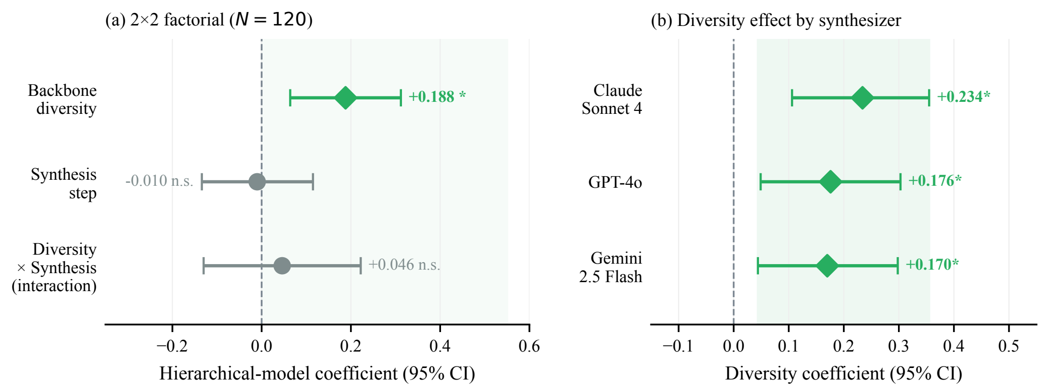


Figure 2: (a) Backbone diversity is the only statistically significant factor in the 2×2 factorial (N=198). The synthesis step produces a null effect (+0.044, CI straddles zero). (b) The diversity coefficient remains positive and significant across all three synthesizer models, confirming the null synthesis effect is not an artifact of synthesizer choice.

primarily on tasks where models make structurally different errors. A variance decomposition (Appendix R) confirms this finding. Between-model variance exceeds within-model variance (B/W ratio ≥ 2) on only two of seven tasks (Circle Packing 15.2, Molecule QED 2.0). On three tasks (TSP-100, MaxCut, DiffBases) within-model noise dominates (B/W < 1), so adding a different backbone contributes no more than resampling the same one.

Coverage, not superiority. To rule out a capability confound, we ran MoA-no-synthesis with same-model teams (GPT-4o$\times$3, Gemini$\times$3) across all nine tasks (n=10 each). On aggregate the diverse team (Q=0.214) and GPT-4o$\times$3 (Q=0.221) score similarly (Table 10, Appendix), so diversity does not make the team *better*. But each same-model team catastrophically fails on at least one task (Q=0.000), while the diverse team never does. The correct framing is *coverage*, not superiority.

Interaction helps same-model teams. Interaction is not uniformly harmful. When all agents share the same backbone, critique-and-revise consistently improves scores. Chain (+0.051), MAgICoRe (+0.044), and Debate (+0.012) all produce positive same-model MIG (Table 6). MoA likewise benefits (+0.012). Same-model interaction refines proposals without overwriting structure, because there is no structural diversity to destroy.

Interaction helps on constraint tasks. Task structure appears to influence whether interaction adds value. On constraint tasks, critique names a specific violation and the refiner patches it without re-solving the whole problem. The first critique round never degrades a constraint task (0/30 runs) but degrades optimization tasks 57% of the time (17/30). MAgICoRe raises feasibility on both constraint tasks with a consistent gradient, from Best-of-N (0%) to Self-Refine (7–13%) to MAgICoRe (47–73%, Fisher exact p<0.003). Full regression rates and feasibility gradients appear in Tables 4–5. On optimization tasks the gradient reverses, with Best-of-N > Self-Refine > MAgICoRe. Self-Refine loses versus Best-of-N on Flat Polynomials (p=0.009, d=1.02) and TSP-100 (p=0.001, d=1.46). The optimization loss replicates on both Gemini and GPT-4o backbones (Appendix E). The constraint win is not a sampling artifact. Best-of-N achieves 0% feasibility at all tested sample sizes because the backbone rarely generates valid solutions on these two tasks independently, so critique provides a qualitatively different path to feasibility. Note that constraint tasks use visible-evaluator feasibility (not hidden scores), so the optimization–constraint comparison involves different evaluation methodologies. This is a limitation of the cross-task comparison.

Table 4: First-round critique regression rate (MAgICoRe, Gemini backbone, n=15 per task). Constraint = Knapsack-50, 3AP-Free-100. Optimization = Erdős, Flat Polynomials.

Task type	Regress step 0→1	Any regression	n
Constraint (2 tasks)	**0/30 (0%)**	5/30 (17%)	30
Optimization (2 tasks)	17/30 (57%)	25/30 (83%)	30

Table 5: Feasibility success rates on two constraint tasks (Gemini backbone, n=15 each, visible evaluator). Fisher exact p compares MAgICoRe to Best-of-N.

Task	Best-of-N	Self-Refine	MAgICoRe	p
Knapsack-50	0%	13%	47%	0.003
3AP-Free-100	0%	7%	73%	$<$0.001

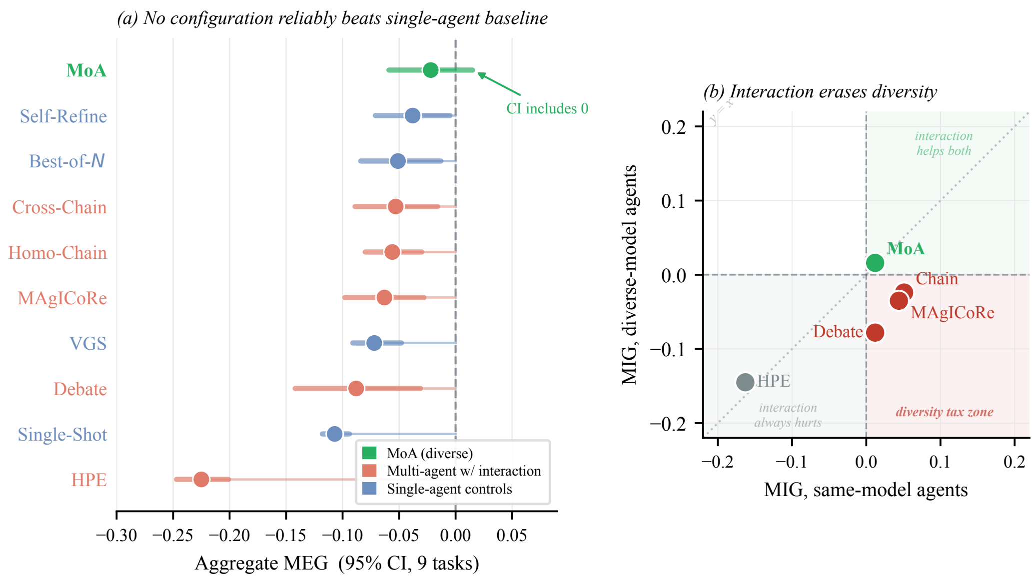


Figure 3: (a) Aggregate MEG for all ten configurations (95% CI, 9 tasks). MoA is the only configuration whose confidence interval includes zero. (b) MIG quadrant scatter (x-axis = same-model MIG, y-axis = diverse-model MIG). MAgICoRe and Debate occupy the *interaction tax zone* (bottom-right quadrant), where interaction helps same-model agents but hurts once backbones are diverse. MoA escapes to the top-right because its proposers never see each other's outputs.

But interaction erases diversity. The positive picture changes when backbones are diverse. Chain, MAgICoRe, and Debate each produce positive MIG with the same-model agents but turn negative under diverse backbones (Table 6, Figure 3b). For MAgICoRe and Debate, the interaction structure is identical across the two backbone conditions, so the loss comes from the diversity it erases. MoA serves as a mechanistic control for this claim. Its proposers use diverse backbones but never see each other's outputs, isolating the effect of diversity from the effect of interaction. MoA's MIG improves from the same model to diverse ($+0.012 \rightarrow +0.016$), confirming that diversity itself is beneficial and that the damage in other configurations comes specifically from the interaction step, not from using different models. Per-task values (Appendix K) confirm the reversal concentrates on Erdős, DiffBases, and MolQED.

Chain caveat. For Chain, backbone composition and chain length covary (4 same-model vs. 3 diverse agents), so the MIG flip confounds diversity with configuration structure. The pattern is consistent with MAgICoRe and Debate but not independently causal.

Aggregate hidden-evaluator scores. No configuration achieves a positive aggregate MEG (Table 7). Diverse MoA is the only configuration whose CI includes zero (-0.021 $[-0.059, +0.014]$). Every configuration where agents exchange intermediate work ranks below Self-Refine (-0.038). MoA ranks above it because its proposers never exchange work. Leave-one-task-out jackknife (Appendix S) confirms sign stability.

Visible and hidden evaluators diverge. On three of nine optimization tasks, visible and hidden evaluators rank configurations differently (Appendix M). The divergences concentrate on open problems without known optimal solutions. On MaxCut, the visible-best configuration (VGS) is not the hidden-best (Spearman $\rho = 0.891$, top-1 inverted). Without the hidden-evaluator split,

Table 6: Mean MIG per configuration family, averaged over eight optimization tasks (TSP-50 excluded for incomplete mixed-model coverage, HPE averages over seven tasks because Erdős data is unavailable). Positive = interaction helped, negative = interaction hurt. $P(\text{same}>\text{div})$: bootstrap probability (10,000 resamples) that same-model MIG exceeds diverse MIG.

Configuration	Same-model MIG	Diverse MIG	$P(\text{same}>\text{div})$
Chain	$+0.051$	-0.024	97%
MAgICoRe	$+0.044$	-0.035	99%
Debate	$+0.012$	-0.078	88%
HPE	-0.163	-0.145	38%
MoA	$+0.012$	$+0.016$	42%

Table 7: Aggregate MEG across nine optimization tasks. 95% CI from 10,000 bootstrap resamples. MoA is the only configuration whose CI includes zero (✓).

Configuration	Type	AggMEG	95% CI	
MoA (diverse)	Multi	-0.021	$[-0.059,\ +0.014]$	✓
Self-Refine	Single	-0.038	$[-0.071,\ -0.005]$	
Best-of-N	Single	-0.051	$[-0.084,\ -0.013]$	
Cross-Chain	Multi	-0.053	$[-0.089,\ -0.016]$	
Homo-Chain	Multi	-0.056	$[-0.080,\ -0.030]$	
MAgICoRe	Multi	-0.063	$[-0.098,\ -0.028]$	
VGS	Single	-0.072	$[-0.091,\ -0.048]$	
Debate	Multi	-0.088	$[-0.142,\ -0.031]$	
Single-Shot	Single	-0.107	$[-0.118,\ -0.094]$	
HPE	Multi	-0.225	$[-0.247,\ -0.201]$	

configurations that query the visible evaluator more effectively would appear superior even if their solutions are no better on the underlying problem.

Mechanism. Step-level trajectories (Appendix V) show that diversity dies in a single round. On Erdős, diverse Debate achieves a strong intermediate score at round 2 but collapses at round 3 once agents read each other's full solutions. After synthesis, pairwise solution distance falls from 0.48 to 0.34, matching same-model MoA (Appendix J). On five of seven tasks, synthesis copies the best proposer's output $\geq 80\%$ of the time rather than combining proposals (Appendix I). Strategy diversity is also limited. GPT-4o outputs the trivial constant on 100% of Erdős runs, so nominal backbone diversity does not guarantee strategy diversity (Appendix T).

5 Discussion and Limitations

The MIG sign reversal (Table 6) holds across three architectures with different interaction structures (Chain, MAgICoRe, Debate), suggesting the mechanism is not architecture-specific but inherent to full output exchange between diverse agents. The 2×2 factorial clarifies why. What matters is whether proposers are different models, not whether a synthesizer aggregates their work. The synthesis coefficient is null ($+0.044$, CI straddles zero), so the standard "combine diverse perspectives" rationale for synthesis does not hold on these tasks. The real value of diversity is coverage. On Erdős, diverse families make structurally different errors, so independent proposals recover what any one missed. But on MolQED, model errors are anti-correlated (Table 21) yet the diversity contrast is only $+0.026$ (Table 19), showing that anti-correlated errors alone do not guarantee a better candidate to select. The benefit requires that at least one model produce a qualitatively different solution, not just a negatively correlated one.

The constraint results (Section 4) reveal when convergence becomes constructive. On constraint tasks, critique targets a specific checkable violation, and the refiner patches it locally without re-solving the problem. On open-ended optimization, critique has no specific flaw to target and instead pushes solutions toward familiar patterns. This distinction suggests a broader principle. Feedback is productive when it is verifiable and local, and destructive when it is holistic and subjective. These findings refine the picture from Jarrett et al. [Jarrett et al., 2025], who found that multi-agent teams uniformly underperform their best member. Our results show this is not universal. The underperformance is specific to configurations where diverse agents read each other's full outputs on open-ended tasks. Architectures that preserve independence (MoA) or tasks that provide verifiable feedback (constraints) escape the pattern.

These results apply to the ten configurations tested here under a specific budget regime. The 2×2 factorial deliberately targets three tasks with meaningful configuration spread to cleanly isolate diversity and synthesis as separable factors. Architectures with coordination before independent search, explicit population diversity, or richer feedback (e.g., code execution) may behave differently. For benchmark design, visible and hidden evaluators can disagree. VGS and HPE, both of which rely on visible-evaluator feedback as a core mechanism, rank among the lowest under hidden evaluation (Table 7), though we cannot claim the bias is systematically directional from three divergent tasks.

The 2×2 factorial was run on three tasks with unequal seeds per condition (n=30 on DiffBases, n=10 on Erdős and MolQED). The diversity coefficient is driven by Erdős (Appendix G), and a MolQED result significant at n=5 dissolved at n=15 (p=0.93). The MIG analysis covers eight tasks, but three (Erdős, DiffBases, MolQED) carry most of the signal, while four produce near-zero MIG. This concentration is consistent with the theory. The interaction tax is predicted to appear only where models make qualitatively different errors (between-model variance dominates), and the variance decomposition confirms this holds on exactly those three tasks. On tasks with correlated errors (ρ=0.43 on TSP-100), diversity adds less. The constraint-boundary finding rests on two tasks and one backbone (Gemini). It may not generalize to other constraint domains or backbones. All configurations run within a single session on mathematical/combinatorial optimization. Designs with coordination before independent search, or tasks requiring simulation or retrieval, fall outside scope.

6 Conclusion

We evaluated ten multi-agent LLM configurations built from three model families on eleven scientific tasks under hidden evaluation. Three findings emerge. First, diverse model families should generate proposals independently. Any architecture that exposes agents to each other's full outputs destroys the diversity that motivated using different models. Second, diversity provides insurance, not improvement. No diverse team outperforms the best same-model team on aggregate, but every same-model team catastrophically fails on at least one task. Third, interaction should be reserved for tasks with verifiable feedback. When critique targets a checkable violation, iterative refinement is productive. When the task is open-ended, critique degrades solutions more often than it improves them. These findings depend on the hidden-evaluator methodology. Visible and hidden rankings diverge on three of nine tasks, meaning that benchmarks relying solely on queryable scorers may produce misleading configuration rankings.

Several directions remain open. Whether coordination *before* independent search, such as planning which subproblems each agent tackles, can recover the benefits of interaction without the convergence cost is untested. Extending this analysis to coding tasks, multi-step reasoning, and retrieval-augmented settings would clarify whether the interaction tax generalizes beyond mathematical and combinatorial optimization. Designs that explicitly maintain population diversity, such as evolutionary approaches with diversity-preserving selection, may offer a path to combining the strengths of diverse models without paying the interaction tax.

Software and Data

All experiments use a frozen testbed snapshot with pinned model identifiers, deterministic verifiers, and logged budget traces. The release artifact includes all task instances, configuration harness, prompt templates, and analysis scripts. An anonymized repository is available at `https://anonymous.4open.science/r/interaction-tax-3244`.

Impact Statement

This work involves no human subjects, private data, or dual-use concerns. The hidden-evaluator design provides a sounder basis for configuration selection than visible-evaluator benchmarks.

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A Full Per-Task MEG

Table 8: Per-task MEG for all ten configurations. MaxCut and LJ-n=41 omitted (all configurations score zero on both tasks).

Task	MoA	SR	BoN	XChain	HChain	MAgI	VGS	Debate	S-Shot	HPE
Circle Packing	-0.238	-0.027	0.000	-0.027	0.000	0.000	0.000	-0.522	0.000	-1.000
Difference Bases	$+0.090$	-0.110	0.000	-0.113	-0.197	-0.145	-0.132	$+0.112$	-0.197	-0.197
Flat Polynomials	$+0.013$	-0.059	-0.077	-0.077	-0.077	-0.077	0.000	-0.038	-0.077	-0.077
TSP-100	-0.015	-0.018	-0.005	-0.005	-0.018	-0.012	0.000	-0.018	-0.009	-0.018
Erdős	$+0.021$	0.000	-0.371	-0.134	-0.048	-0.149	-0.476	-0.334	-0.476	-0.476
Molecule QED	-0.013	-0.049	0.000	-0.102	-0.122	-0.137	-0.041	$+0.024$	-0.110	-0.240
TSP-50	-0.050	-0.077	-0.003	-0.021	-0.038	-0.046	0.000	-0.018	-0.090	-0.015

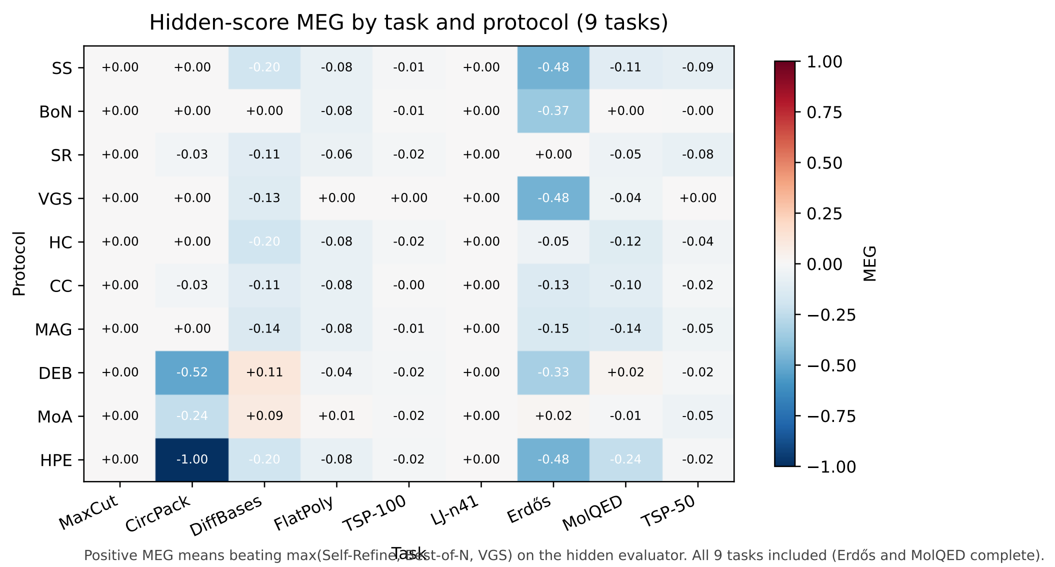

Figure 4: Per-task MEG for all ten configurations on nine optimization tasks under hidden evaluation. Three tasks (Erdős, DiffBases, MolQED) account for nearly all inter-configuration variation. MaxCut and LJ-n=41 produce zero signal.

B Per-Backbone Coverage

Table 9: Per-backbone Best-of-N Q-scores on six tasks where backbone scores clearly diverge. Bold = best backbone. Each model is the top scorer on at least one task. MolQED omitted (Claude and GPT-4o tie). — = not run (GPT-4o Best-of-N was not collected on TSP-50).

Task	Claude	GPT-4o	Gemini	Winner
Circle Packing	**1.000**	0.519	0.745	Claude
TSP-50	**0.141**	—	0.115	Claude
Erdős	0.131	**0.710**	0.000	GPT-4o
TSP-100	0.013	**0.021**	0.010	GPT-4o
Difference Bases	0.209	0.264	**0.610**	Gemini
Flat Polynomials	0.000	0.007	**0.143**	Gemini

C Capability Confound Check

Table 10: MoA-no-synthesis aggregate Q-scores across nine optimization tasks (n=10 per condition). The diverse team and GPT-4o×3 have similar point estimates, but each same-model team catastrophically fails on at least one task (Q=0.000), while the diverse team never does.

Configuration	Aggregate Q	Min task Q
Diverse (Claude + GPT-4o + Gemini)	0.214	> 0
GPT-4o×3	0.221	0.000
Gemini×3	0.157	0.000

D Within-Backbone Comparisons (n=15)

Table 11 reports all within-backbone comparisons (n=15 per condition unless noted). The molecule-QED result that appeared significant at n=5 (p=0.046) dissolved at n=15 (p=0.93), illustrating that five seeds per condition is marginal for within-backbone effect detection. In the MolQED row the BoN column retains the original n=5 because the comparison is against the initial result that motivated the replication. The n=15 MAgICoRe runs showed no effect regardless of baseline sample size.

Table 11: Within-backbone comparisons (Gemini, n=15 per condition unless noted). Positive d = iterative configuration is worse (higher score on a minimize task or lower on a maximize task).

Task	Dir	Configuration	n	Iter. mean	BoN mean	d	p
Flat Poly	min	MAgICoRe-G	15 vs 15	2.087	1.899	+1.25	0.002
Flat Poly	min	Self-Ref-G	15 vs 15	2.020	1.899	+1.02	0.009
TSP-100	min	MAgICoRe-G	15 vs 15	53,351	50,786	+1.42	0.001
TSP-100	min	Self-Ref-G	14 vs 15	53,239	50,786	+1.46	0.001
TSP-50	min	MAgICoRe-G	15 vs 15	24,167	23,103	+1.23	0.002
Molecule QED	max	MAgICoRe-G	15 vs 5	0.863	0.859	+0.05	0.93

E Optimization Loss Across Backbones

The optimization loss pattern replicates on both tested backbones (Table 12). Gemini and GPT-4o both show significant degradation from iterative refinement on Flat Polynomials ($d \approx 1.3$) and TSP-100 ($d = 1.4$–2.6).

Table 12: MAgICoRe vs. Best-of-N on two optimization tasks, two backbones (n=15 each). The loss replicates on both tested backbones.

Task	Backbone	MAgICoRe	Best-of-N	d	p
Flat Poly	Gemini	2.087	1.899	+1.25	0.002
Flat Poly	GPT-4o	2.951	2.251	+1.30	0.001
TSP-100	Gemini	53,351	50,786	+1.42	0.001
TSP-100	GPT-4o	53,529	50,128	+2.64	<0.001

F Constraint Feasibility Gradient

Within the Gemini backbone (n=15 each), constraint feasibility generally increases with configuration sophistication (Table 13). Critique-based configurations outperform evolutionary search on both tasks. Single-Shot outperforms Best-of-N on Knapsack (7% vs. 0%), breaking a strict monotonic gradient at the sampling end.

Table 13: Constraint feasibility rates by configuration type (Gemini backbone, n=15). Critique outperforms evolution on both tasks. The sampling end is not monotonic (Single-Shot > BoN on Knapsack).

Task	Single-Shot	Best-of-N	VGS (evolutionary)	MAgICoRe (critique)
Knapsack-50	7%	0%	27%	47%
3AP-Free-100	0%	0%	13%	73%

G Leave-One-Out Sensitivity

Table 14: Leave-one-out sensitivity of the diversity main effect. Only removing Erdős drops the coefficient to near zero.

Task withheld	$\hat{\beta}_{\text{div}}$	95% CI	p
None (all 3 tasks, N=198)	+0.195	[+0.125, +0.262]	<0.001
Erdős withheld (n=158)	+0.052	[−0.013, +0.117]	0.10
DiffBases withheld (n=80)	+0.254	[+0.144, +0.358]	<0.001
MolQED withheld (n=158)	+0.181	[+0.097, +0.266]	<0.001

H Cross-Synthesizer Replication

Table 15: Diversity coefficient under three synthesizer backbones. The coefficient remains positive and significant for all three. The synthesis coefficient remains near zero.

Synthesizer	$\hat{\beta}_{\text{div}}$	95% CI
Claude Sonnet 4	$+0.234$	$[+0.106,\ +0.355]$
GPT-4o	$+0.176$	$[+0.049,\ +0.303]$
Gemini 2.5 Flash	$+0.170$	$[+0.044,\ +0.298]$

I Synthesis Decision Quality

Table 16: Synthesis output classified relative to the best-scoring proposer. On five of seven tasks synthesis copies the best proposal $\geq$80% of the time. On Difference Bases it degrades 50% of the time.

Task	Improved	Copied	Degraded
Circle Packing	60%	40%	0%
Molecule QED	20%	80%	0%
Erdős	10%	80%	10%
Flat Polynomials	0%	100%	0%
TSP-100	0%	100%	0%
TSP-50	0%	90%	10%
Difference Bases	7%	43%	50%

J Diversity Collapse

Table 17: Mean pairwise solution distance (Hamming for binary tasks, cosine for continuous). After synthesis, diverse MoA distance (0.48) falls to 0.34, matching same-model MoA (0.34).

Configuration	Avg pairwise distance
MoA (no synthesis)	0.48
Best-of-N (same model)	0.40
Debate	0.38
MoA (diverse + synthesis)	0.34
MoA (same-model)	0.34

K Per-Task MIG Breakdown

Table 18: Marginal Interaction Gain (MIG) per task for five configuration families (eight optimization tasks, TSP-50 excluded due to incomplete mixed-model coverage). Bold marks the Erdős sign flips, where the reversal is largest. Several smaller sign flips also occur on DiffBases and MolQED. HPE Erdős is marked because no HPE runs were collected on that task. On tasks with no Q-score variance (MaxCut, LJ-n41), MIG is zero for all configurations.

Family	Cfg	MCut	CPack	DBases	FPoly	TSP1	LJ41	Erdős	MolQ
MAgICoRe	homo	$+0.000$	$+0.000$	$+0.053$	$+0.000$	-0.002	$+0.000$	$\mathbf{+0.327}$	-0.027
MAgICoRe	mixed	$+0.000$	$+0.000$	-0.072	-0.002	-0.006	$+0.000$	$\mathbf{-0.284}$	$+0.088$
Debate	homo	$+0.000$	-0.522	$+0.309$	$+0.040$	-0.009	$+0.000$	$+0.142$	$+0.134$
Debate	mixed	$+0.000$	-0.602	-0.140	$+0.018$	$+0.000$	$+0.000$	$+0.142$	-0.039
HPE	homo	$+0.000$	-1.000	$+0.000$	$+0.000$	-0.009	$+0.000$	—	-0.130
HPE	mixed	$+0.000$	-0.700	-0.056	-0.034	$+0.006$	$+0.000$	—	-0.227
Chain	homo	$+0.000$	$+0.000$	$+0.000$	$+0.000$	-0.009	$+0.000$	$\mathbf{+0.428}$	-0.012
Cross-Chain	mixed	$+0.000$	-0.027	-0.056	-0.034	$+0.005$	$+0.000$	$\mathbf{-0.084}$	$+0.008$
MoA	homo	$+0.000$	-0.077	$+0.071$	$+0.004$	-0.009	$+0.000$	$+0.071$	$+0.038$
MoA	mixed	$+0.000$	-0.238	$+0.148$	$+0.056$	-0.006	$+0.000$	$+0.071$	$+0.098$

L Per-Task 2×2 Cell Means

Table 19: Mean Q-scores for each arm of the 2×2 diversity $\times$ synthesis factorial, broken down by task (n=10 per cell for Erdős and MolQED, $n\approx30$ for DiffBases). The diversity contrast (mean of diverse arms minus mean of same-model arms) shows that Erdős drives the aggregate diversity coefficient, DiffBases contributes moderately, and MolQED contributes near zero. The synthesis contrast is near zero on all three tasks.

Task	Same+Sel	Same+Syn	Div+Sel	Div+Syn	Div contrast	Syn contrast
DiffBases	0.202	0.071	0.237	0.287	+0.126	−0.041
Erdős	0.031	0.071	0.568	0.497	+0.482	−0.016
MolQED	0.314	0.377	0.307	0.437	+0.026	+0.097

M Visible–Hidden Rank Agreement

Table 20: Spearman rank correlation between visible-evaluator and hidden-evaluator configuration rankings per task (n=10 configurations). Three tasks show $\rho < 1$. On MaxCut, the visible-best configuration (VGS) is not the hidden-best (Cross-Chain). On six of nine optimization tasks, visible and hidden rankings agree perfectly.

Task	Spearman ρ	Top-3 overlap	Top-1 inverted?
MaxCut	0.891	2/3	Yes
Circle Packing	1.000	3/3	No
DiffBases	0.806	2/3	No
Flat Poly	1.000	3/3	No
TSP-100	1.000	3/3	No
LJ-n=41	1.000	3/3	No
Erdős	1.000	3/3	No
MolQED	0.952	3/3	No
TSP-50	1.000	3/3	No

N Inter-Model Error Correlation

Table 21: Spearman rank correlations of per-seed hidden scores between model pairs on four tasks with nonzero cross-model Q-score variance (Erdős, DiffBases, MolQED, TSP-100). Low or negative ρ indicates uncorrelated errors (diversity helps). Positive ρ indicates correlated errors (diversity adds less). Gemini is constant on Erdős (all seeds produce raw hidden score 0.503, which maps to Q=0.000 after normalization), precluding correlation.

Task	Claude–GPT-4o ρ	Claude–Gemini ρ	GPT-4o–Gemini ρ	Error independence
Erdős	+0.18	— (constant)	— (constant)	Yes — uncorrelated
DiffBases	+0.05	+0.12	+0.17	Yes — uncorrelated
MolQED	−0.45	−0.20	−0.33	Yes — anti-correlated
TSP-100	+0.38	+0.50	+0.41	No — correlated

O Transcript Evidence

Two examples illustrate how diversity is preserved or wasted depending on whether agents interact before the final decision.

Example 1. Synthesis selects the wrong proposal (MoA, Erdős, seed 1). Three proposers generate independently. Claude produces a structured binary-pattern function (devScore = 0.490). GPT-4o outputs a trivial uniform array of constant 0.005 values (devScore = 0.250). The synthesizer receives both, correctly identifies Claude's higher score, then reasons as follows.

> "Agent 2 uses a uniform distribution with constant value 0.005 across all points. This achieves exactly the theoretical lower bound of $1/4$ for uniform distributions. . . The uniform approach is mathematically optimal. . . Agent 2's solution is superior as it achieves the lower bound with perfect consistency."

The synthesizer constructs a plausible-sounding but incorrect mathematical justification for choosing the worse solution. The final output scores 0.250, discarding the 0.490 solution that diversity produced. Diversity existed, the synthesizer saw it, and still made the wrong choice.

Example 2. Synthesis selects the right proposal (MoA, MolQED, seed 1). Three proposers produce structurally distinct molecules. Claude produces `COc1ccc(Cc2cnc(N)nc2N)cc1` (methoxybenzene

+ pyrimidine, devScore = 0.828). GPT-4o produces an aspirin derivative (devScore = 0.685). Gemini produces a small acetyl amino acid (devScore = 0.547). The synthesizer identifies Claude's molecule as containing "a methoxybenzene ring (drug-like aromatic system) and a pyrimidine ring with amino groups (common in pharmaceuticals)" and correctly selects it. Here the quality gap is large ($0.83 \gg 0.55$) and the solution format is chemically interpretable. Synthesis works when the best proposal's advantage is obvious.

Pattern. These two examples bracket the synthesis mechanism. On Erdős, the best proposer's advantage is encoded in a numerical array with no interpretable structure, and the synthesizer fabricates a reason to reject it. On MolQED, the best proposer's advantage is visible in molecular structure, and the synthesizer identifies it correctly. Synthesis is reliable only when the quality signal is large and the format is interpretable. This is consistent with the null synthesis coefficient in the 2×2 factorial (Table 3), where across tasks these two failure modes cancel out.

P Step-Level Solution Distance

Table 22: Pairwise solution distance between consecutive steps within each run, averaged over seeds. On Debate diverse, distance drops sharply after agents exchange outputs (round 2→3), confirming diversity collapse in real time. Same-model Debate shows no comparable drop. Metrics are Jaccard distance for DiffBases (discrete sets), cosine distance for Erdős (continuous arrays), and Levenshtein distance for MolQED (SMILES strings).

Task	Configuration	Step 1→2	Step 2→3	Pattern
DiffBases	Debate (same)	0.950	0.891	Slight convergence
	Debate (diverse)	0.956	**0.479**	Collapse at round 3
	MAgICoRe (diverse)	0.953	0.815	Convergence
Erdős	Debate (same)	0.500	—	Single change
	Cross-Chain	0.128	0.199	Low throughout
	MAgICoRe (diverse)	0.448	0.293	Shrinking
MolQED	Debate (same)	0.571	0.412	Converging
	Debate (diverse)	0.669	**0.378**	Fast convergence

Q Proposer Score Spread

Table 23: Mean pairwise score spread among MoA proposers within each run, measuring how different the proposals are before synthesis. Diverse proposers produce 2.2× more score spread than same-model proposers on Erdős. On DiffBases, within-model variance dominates, so backbone diversity adds less spread. Diversity is real at proposal time. Synthesis erases it (Section 5.4).

Task	Diverse spread	Same-model spread	Ratio
Erdős	0.448	0.207	2.2×
MolQED	0.222	0.152	1.5×
DiffBases	1500	3045	0.5×

R Within-Model vs. Between-Model Variance

Table 24: Variance decomposition of single-shot Q-scores across three backbones (n=5 seeds each). Between-model variance (σ_B^2) measures how different the model means are. Within-model variance (σ_W^2) measures per-seed noise. A B/W ratio above 1 indicates that models are more different from each other than from themselves, the prerequisite for backbone diversity to help. Two tasks clearly meet this criterion ($B/W \geq 2$) and two are marginal ($B/W \approx 1.2$–1.3). LJ-n=41 and TSP-50 are omitted (LJ because all models score Q=0, TSP-50 for incomplete backbone coverage).

Task	σ_W^2	σ_B^2	B/W	Diversity useful?
Circle Packing	0.017	0.253	15.2	Yes
MolQED	0.015	0.030	2.0	Yes
Flat Poly	0.054	0.070	1.3	Marginal
Erdős	0.050	0.061	1.2	Marginal
TSP-100	0.020	0.010	0.5	No
MaxCut	0.0001	0.00002	0.3	No
DiffBases	0.095	0.016	0.2	No

S Jackknife MEG Stability

Table 25: Leave-one-task-out jackknife of aggregate MEG for the ten core configurations. Each column shows AggMEG recomputed after dropping one task. Sign-stable = AggMEG sign never flips across all nine leave-one-out subsets. Nine of ten core configurations are sign-stable. MoA flips positive only when Circle Packing (its worst task) is dropped. Across all 49 configuration variants in the benchmark, 46 (94%) are sign-stable.

Configuration	AggMEG	JK min	JK max	Sign-stable?
MoA	-0.021	-0.035	$+0.006$	No
Self-Refine	-0.038	-0.042	-0.029	Yes
Best-of-N	-0.051	-0.057	-0.011	Yes
Cross-Chain	-0.053	-0.060	-0.043	Yes
Homo-Chain	-0.056	-0.062	-0.038	Yes
MAgICoRe	-0.063	-0.071	-0.052	Yes
VGS	-0.072	-0.081	-0.022	Yes
Debate	-0.088	-0.113	-0.034	Yes
Single-Shot	-0.107	-0.120	-0.060	Yes
HPE	-0.225	-0.253	-0.128	Yes

T Model Strategy Classification

Table 26: Strategy frequency per backbone on Erdős and DiffBases, classified from rawOutput keywords across all single-shot seeds. Categories are non-exclusive keyword tags, so counts may exceed the number of seeds. GPT-4o outputs a trivial constant (≈ 0.005) on 100% of Erdős runs. Gemini also produces near-trivial constants (all zeros, raw score ≈ 0.503). Only Claude generates structurally varied, non-trivial solutions. On DiffBases, GPT-4o uses triangular numbers on every run.

Task	Strategy type	Claude	GPT-4o	Gemini
Erdős	Trivial constant	0/10	**5/5**	**5/5**
	Binary/block pattern	4/10	0/5	0/5
	Smooth/cosine-based	2/10	0/5	0/5
	Mathematical reasoning	6/10	0/5	0/5
DiffBases	Sidon sets (B_2 sets)	4/10	0/5	1/5
	Triangular numbers	1/10	**5/5**	1/5
	Arithmetic progression	1/10	0/5	2/5
	Greedy/algorithmic	2/10	0/5	0/5

On Erdős, GPT-4o and Gemini each produce a single stereotyped output across all seeds, contributing zero strategy variance to any diverse ensemble. A "three-backbone" team has effectively one contributor. On DiffBases, GPT-4o is similarly fixed (triangular numbers on 100% of runs), though Gemini and Claude each use multiple strategies. The effective diversity of an ensemble depends on whether each backbone explores different regions of the solution space, not merely on the number of distinct model families.

This has a direct consequence for the MIG results. When a diverse Debate or Cross-Chain team includes GPT-4o on Erdős, one agent always proposes the trivial constant ($Q{=}0$). The interaction step then risks adopting this trivial proposal, producing the regression visible in Table 27.

U Additional Transcript Examples

Three further examples complement the synthesis examples in Appendix O. Together they illustrate the diversity bottleneck (Example 3), homogenization-as-improvement (Example 4), and the positive case where independent generation preserves diversity (Example 5).

Example 3. Cross-Chain on Erdős (diversity bottleneck, seed 1). Step 1 (Claude, devScore $= 0.500$). Claude reasons about minimizing overlap with translates and produces a non-trivial structured function. Step 2 (GPT-4o, devScore $= 0.250$). GPT-4o sees Claude's solution but outputs a trivial constant $f = 0.5$ (GPT-4o defaults to a near-zero constant when generating independently, and in chain context it shifts to 0.5, but both are trivial and score identically). Step 3 (Gemini, devScore $= 0.694$). Gemini ignores GPT-4o's trivial answer and produces a strong binary pattern. The run's final hidden score is 0.250 because the configuration selected GPT-4o's trivial answer. When a weak model sits in the middle of a chain, it acts as a *diversity bottleneck*, replacing a good solution with a trivial one.

Example 4. Debate diverse on DiffBases (homogenization improves score, seed 1). Round 1 (devScore $= 597$). The model reasons creatively about fractal-like structures for a difference basis set. Round 2 (devScore $= 157$). After seeing the first model's approach, the agent shifts to a standard triangular-number construction. Hidden score is 70.2. This is a positive example of interaction. The solution improved by replacing a creative but poorly executed approach with a conventional mathematical construction. But the improvement came from homogenization. The solution got better by becoming less creative. This is exactly the mechanism the diversity coefficient captures at aggregate, where interaction rewards convergence toward standard approaches.

Example 5. MoA-nosynth on Erdős (diversity preserved, seed 1). Three proposers generate independently. Claude scores 2.47 (structured sparse function). GPT-4o scores 0.34 (compact near-constant solution). Gemini scores 0.50 (different interpolation). The configuration picks the best (GPT-4o's 0.34), which is better than any Claude-only attempt on the same seed (best of three is 0.392). The diverse ensemble finds a solution Claude does not naturally produce. The gain is real, but only because the agents generated without seeing each other. The moment they interact (Example 3 above), the same team loses this advantage.

V Convergence Mechanism Details

Step-level trajectories reveal when diversity dies. On Erdős, diverse Debate achieves a strong intermediate score at round 2, when agents exchange only critiques, but collapses at round 3 once agents read each other's full solutions (Table 27). The damage occurs at the moment of full output exchange, not gradually. MAgICoRe shows task-dependent patterns (Table 28). TSP-100 improves slightly through both rounds, Flat Polynomials degrades monotonically, Molecule QED peaks at step 1 then reverts, and Erdős converges to a trivial solution. Step-level trajectories appear in Figure 5.

On five of seven tasks, synthesis copies the best proposer's output $\geq$80% of the time (Table 16). On Erdős, the synthesizer selects a trivial constant (visible score 0.250) over a structured function (visible score 0.490), constructing a plausible but incorrect justification (Appendix O). Strategy diversity is also limited. GPT-4o outputs the trivial constant on 100% of Erdős runs, and Gemini defaults to the same trivial solution (Table 26). On that task a three-backbone team effectively has one non-trivial contributor (Claude), so nominal backbone diversity does not guarantee strategy diversity.

Diverse MoA proposers produce $2.2\times$ more score spread than same-model proposers on Erdős, though the ratio is task-dependent ($1.5\times$ on MolQED, $0.5\times$ on DiffBases, Table 23). After synthesis, pairwise solution distance falls from 0.48 to 0.34, matching same-model MoA (0.34, Table 17). On DiffBases, Debate diverse drops from Jaccard distance 0.956 to 0.479 in a single round (Table 22).

W Per-Round Debate Scores

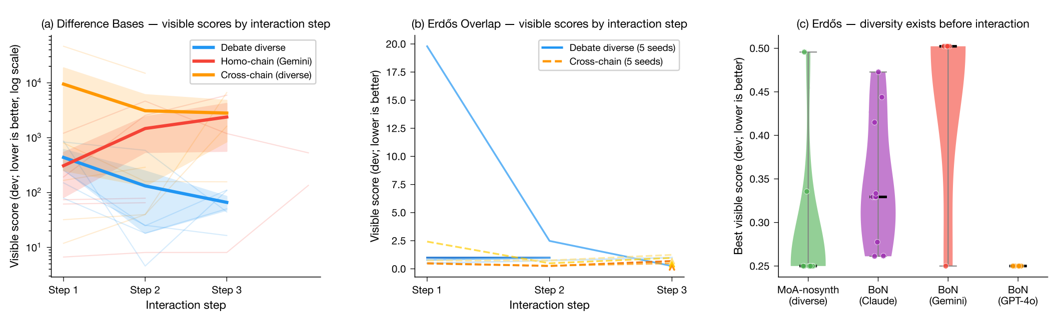

Figure 5: Per-round visible-score trajectories on two optimization tasks (lower is better). These traces apply the visible/dev evaluator to each intermediate artifact. The hidden evaluator is run only once on the final selected submission. (a) DiffBases, where sequential interaction often improves at step 2 but loses that advantage by the final step (faded lines = individual seeds, bold = means). (b) Erdős, where diverse interaction improves at round 2 then regresses at round 3 after agents exchange full outputs. (c) Erdős, where diverse MoA-nosynth (no interaction) achieves broader proposal-time coverage than same-backbone Best-of-N, confirming that diversity exists before interaction and is then erased by it.

Table 27: Per-round visible scores for same-model and diverse Debate on two tasks (lower is better for both, bold = best round). Same-model Debate improves monotonically. Diverse Debate reverses direction at round 3 after agents have exchanged their full outputs.

Configuration	Task	Round 1	Round 2	Round 3
Debate (same)	DiffBases	436	132	**66**
Debate (diverse)	DiffBases	889	**72**	274
Debate (same)	Erdős	4.61	1.24	**0.62**
Debate (diverse)	Erdős	1.28	**0.30**	0.38

X MAgICoRe Step-by-Step Trajectories

Table 28: Mean devScore at each MAgICoRe step (step 0 = initial attempt, step 1 = after first critique, step 2 = after second critique), averaged over n=15 seeds. Direction arrows indicate whether lower (↓) or higher (↑) is better. On optimization tasks, critique frequently degrades the solution after step 1 (Mol. QED) or has negligible effect (TSP-100). *Only 2/15 Erdős seeds reach step 2.

Task	Dir	Step 0	Step 1	Step 2	Trend	Note
TSP-100	↓	54,054	53,686	53,638	Slight improve	Still below BoN-Gemini baseline
Flat Poly	↓	2.290	2.297	2.345	Degrading	Critique changes values, not structure
Mol. QED	↑	0.731	**0.830**	0.785	Peak then revert	Step 1 gains lost at step 2
Erdős	↓	0.699	0.504	—*	Converges to trivial	12/15 seeds drop to 0.250 (constant f)

On optimization tasks, critique frequently degrades solutions. Molecule QED peaks at step 1 (+13.5%) but reverts at step 2. On Erdős, 12 of 15 seeds converge to the trivial constant function at 0.250. The critic steers Gemini toward the simplest possible answer, which scores better on the visible evaluator but remains at Q=0 on hidden evaluation. Flat Poly shows no structural change across steps, so critique changes coefficient values without learning sparsity. These trajectories are the per-run manifestation of the 0/30 vs. 17/30 first-round regression rate in Table 4.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state three claims. (1) Interaction erases diversity within one round. (2) Constraint tasks are an exception where interaction helps. (3) A hidden-evaluator methodology reveals visible/hidden ranking divergence. Each is supported by convergence analysis (Section 4), constraint results (Section 4), and rank-agreement analysis (Section 4 and Appendix), respectively. Limitations on task count, budget regime, and leave-one-out sensitivity are stated in Section 5.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 discusses the following limitations. The 2×2 factorial covers only three tasks. The diversity coefficient is driven by Erdős (leave-one-out analysis in Appendix G). The constraint finding rests on two tasks and one backbone. All configurations run within a single session. Architectures with coordination before independent search are untested.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

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4. **Experiments**

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   Answer: [Yes]

   Justification: Section 3 specifies all ten configurations with pseudocode-level detail (Table 2), the three model families and their identifiers, the eleven tasks, seed counts, and the hidden/visible evaluation split. The Software and Data section describes the release artifact.

5. **Open access to data and code**

   Question: If you ran experiments, did you include the code, data, and instructions needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: An anonymized repository is provided at the URL in the Software and Data section. It contains all task instances, configuration harness, prompt templates, saved results (1556 JSON files), analysis scripts, and figure-generation scripts. Instructions for verifying saved scores and reproducing all tables and figures are in the repository README.

6. **Experimental setting/details**

   Question: Did you specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen)?

   Answer: [Yes]

   Justification: No models are trained. All experiments use frozen API-based LLMs. Section 3 specifies the budget vector (T=200K tokens, W=600s, C=30s, K=25 evaluator calls), seed counts per condition (n=5 core, n=10–30 for the 2×2 factorial, n=15 for within-backbone follow-ups), model identifiers, and temperature settings. Configuration mechanics are detailed in Table 2.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All aggregate results include 95% bootstrap confidence intervals (Tables 3, 7). Per-task results are reported individually. The 2×2 factorial reports coefficient estimates with CIs. Effect sizes (d) and p-values are reported for within-backbone comparisons (Table 10). Leave-one-out jackknife confirms sign stability (Appendix S).

8. **Compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources used?

Answer: [Yes]

Justification: All experiments use API-based LLM inference (Claude Sonnet 4, GPT-4o, Gemini 2.5 Flash) via OpenRouter. Token budgets are logged per run. No custom training or GPU compute is required.

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    Justification: LLMs are the experimental subjects. Section 3 identifies the three model families (Claude Sonnet 4, GPT-4o, Gemini 2.5 Flash), their provider (OpenRouter), and the exact role each plays in every configuration. All prompts are included in the release artifact.